LipScribe presenter guide

Group 4 proposal presentation. For each slide: what to say, the background behind it, where each number comes from, and the questions the lecturer is likely to ask. Read the background until you can explain it without the page. The requirements state that any member may be questioned on any part of the project.

HOW THE TALK IS MARKED

The proposal is worth 10 marks, in five criteria of 2 marks each. Know which slides earn which marks.

Criterion (2 marks each)	What the lecturer looks for	Slides
Real problem, stakeholder, significance	A meaningful problem tied to an identifiable user or decision	2, 3
Precise CV task and scope	Clear input and output, correct task type, substantial but feasible	3, 5
Credible dataset and plan	Data exists or is obtainable; split, annotation and risks considered	4
Architecture, baseline, advanced components	Coherent pipeline, sensible baseline, transfer learning, two justified advanced capabilities	5, 6, 7
Evaluation, feasibility, risks, plan	Metrics fit the task, risks are realistic, the weekly plan produces evidence	8, 9, 10

TIMING

Eight minutes of talk, then about five minutes of questions. The times beside each slide add up to 8:00. The pipeline slide gets the most time because it carries the most marks. Slides 11 (References) and 12 (Thank You) need no talking.

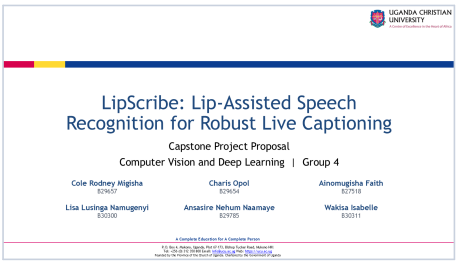
THE ONE-SENTENCE VERSION

Captioning tools that only listen fail in noise. Lips are not affected by noise. LipScribe reads the lips from video and fuses that with the audio, so captions stay usable when the room is loud. We train on the GRID corpus with added noise and measure word error rate at five noise levels against three baselines.

AGREE THESE AS A GROUP BEFORE YOU PRESENT

These points are on the slides but are not in the written proposal. Make sure everyone accepts them, or change the slides.

- **Stretch goals.** Slide 9 treats emotion detection and language identification as stretch goals. The proposal lists them as two of four objectives.
- **Self-recorded clips.** Slides 9 and 10 promise a test on short clips you record yourselves.
- **Transfer-learning steps.** Freeze, then unfreeze, then compare (slide 6) is a plan the proposal does not spell out.
- **What the prototype will not do.** The four limits on slide 3.
- **The six risks and mitigations** on slide 9, and the weekly plan on slide 10.



SLIDE 1 | 0:20

Title

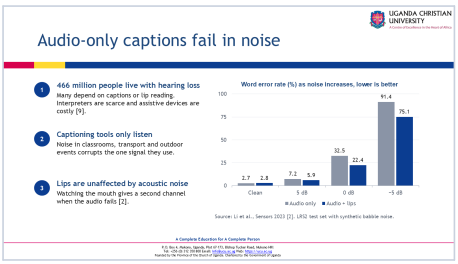
WHAT TO SAY

- We are Group 4, presenting LipScribe.
- LipScribe improves live captions in noisy places by watching the speaker's lips as well as listening.
- We will cover the problem, the user, the data, the pipeline, our baselines, two advanced components, evaluation, risks and the six-week plan.

BACKGROUND YOU SHOULD UNDERSTAND

Keep this short. The requirements say to begin with the problem and the required output, not with the name of a model, so move to slide 2 quickly and do not mention 3D-CNNs or attention here.

The name: "lip-assisted" is deliberate. The system is not lip reading alone. It is audio speech recognition that is helped by lip reading when the audio is poor.



SLIDE 2 | 0:55

Audio-only captions fail in noise

WHAT TO SAY

- 466 million people live with hearing loss. Many rely on captions or lip reading, and interpreters and assistive devices are scarce or costly.
- Today's captioning tools only listen, so they break down in noisy classrooms, transport and outdoor events.
- Read the chart: audio-only goes from 2.7% word error rate in clean audio to 32.5% at 0 dB. Adding lips brings that to 22.4%.
- Lips are not affected by acoustic noise. That is the opening for computer vision.
- At minus 5 dB both systems are still poor, so we will report every noise level honestly.

BACKGROUND YOU SHOULD UNDERSTAND

Automatic speech recognition (ASR) converts spoken audio into text. It is what powers live captions in meeting and phone apps. It works well in quiet rooms and degrades when noise, overlapping speakers or muffled speech corrupt the signal.

Word error rate (WER) is the standard measure. Compare the system's text with the correct text and count substitutions (S), deletions (D) and insertions (I). WER = (S + D + I) / N, where N is the number of words in the correct text. Lower is better. 32.5% means roughly one word in three is wrong, which makes a caption hard to follow. WER can exceed 100% if the system inserts many extra words.

Signal-to-noise ratio (SNR), in decibels, says how strong the speech is compared with the noise. At 10 dB the speech has ten times the power of the noise. At 0 dB they are equal. At minus 5 dB the noise has about three times the power of the speech. Lower SNR means a noisier recording.

Babble noise is the sound of many people talking at once. It is the hardest noise for speech recognition because it has the same character as the speech you want.

Why the chart matters. In clean audio the lips add nothing (2.7% against 2.8%). The gain appears only as noise rises, and it grows with the noise. That is exactly the claim of the project: vision is a second channel that matters when the first one fails.

NUMBERS ON THIS SLIDE AND WHERE THEY COME FROM

Number	What it is	Source
466 million	People worldwide with hearing loss	Kumar et al. 2022 [9]
2.7% / 2.8%	WER in clean audio: audio only / audio plus lips	Li et al. 2023 [2], Table 5
32.5% / 22.4%	WER at 0 dB babble noise: audio only / audio plus lips	Li et al. 2023 [2], Table 5
91.4% / 75.1%	WER at minus 5 dB: audio only / audio plus lips	Li et al. 2023 [2], Table 5

Watch out: The chart is from LRS2 (open television speech). Our own target of below 10% WER is on GRID, which has a small fixed vocabulary and is much easier. Never compare the two sets of numbers directly.

QUESTIONS YOU MAY BE ASKED

Why is this a computer-vision problem rather than a speech problem?

The information that survives noise is visual. The new work in this project is reading a video of the mouth: locating it, encoding its shape and motion, and turning a sequence of frames into text. The audio side is the baseline we are trying to beat.

Those numbers are someone else's results. What are yours?

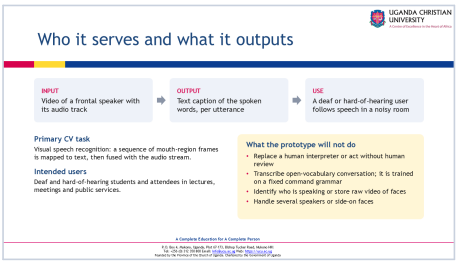
Correct, the chart is published evidence that the problem is real and that lips help. It was measured on LRS2, a dataset of BBC television speech. We have no results yet. Our own numbers come from Week 2 onward, on GRID.

At minus 5 dB the fused system still gets three words in four wrong. Is that useful?

No, and we will say so in the report. The realistic operating range is around 0 to 5 dB, where the gain is large enough to matter. Reporting the level where it fails is part of an honest evaluation.

Is this a local problem?

Noisy lecture rooms and a shortage of interpreters are real conditions on our own campus. The data, however, is not local, which we list as a limitation on slide 9.



SLIDE 3 | 0:45

Who it serves and what it outputs

WHAT TO SAY

- User: a deaf or hard-of-hearing person following speech in a lecture, meeting or public service.
- Input: video of a frontal speaker with its audio. Output: a text caption per utterance.
- Use: the person can follow what is said in a noisy room.
- Primary CV task: visual speech recognition, a sequence of video frames mapped to text, fused with audio.
- It does not replace an interpreter, handle open conversation, identify speakers, or cope with several speakers or side-on faces.

BACKGROUND YOU SHOULD UNDERSTAND

The requirements ask for an explicit **input, output, use** statement, and for what the system will **not** do. This slide exists to answer those directly.

Visual speech recognition (VSR) is the technical name for lip reading by a computer. **Audio-visual speech recognition (AVSR)** uses both sound and video. LipScribe is AVSR.

Why this is a sequence task. The familiar CV tasks give one answer per image: a class label (classification), boxes (detection) or a mask (segmentation). Here the input is a sequence of frames and the output is a sequence of characters of a different length. That is why the model needs a sequence component (BiLSTM) and a sequence loss (CTC), covered on slide 5.

Why each limit is there. GRID has one frontal speaker per clip and a 51-word vocabulary, so the prototype cannot claim open conversation, several speakers or side-on faces. It does not identify people because that would turn a captioning aid into a surveillance tool. The requirements also say a project must not claim autonomous decision authority where human review is still needed.

QUESTIONS YOU MAY BE ASKED

Why is text the right output for this user?

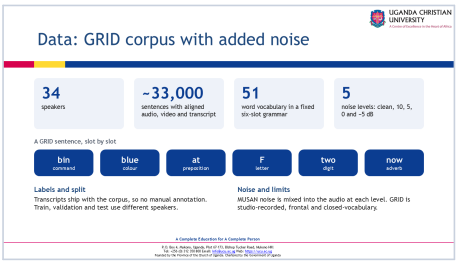
The user needs the words. A class label or a bounding box would not let them follow speech. A caption does.

If it only knows 51 words, how does it help a real deaf student?

As a prototype it does not, and we do not claim that. It demonstrates the method: that fusing lips with audio keeps error rates lower in noise. Moving to natural speech needs a larger dataset such as LRS2 and is future work.

Who is the stakeholder who would act on this?

The end user is the deaf or hard-of-hearing person. An institution's disability support or teaching office would be the one to deploy it in lecture rooms.



SLIDE 4 | 0:50

Data: GRID corpus with added noise

WHAT TO SAY

- We use the GRID audiovisual corpus: 34 speakers, about 33,000 short sentences, clean frontal video with aligned audio.
- Every sentence follows a fixed six-slot grammar, for example "bin blue at F two now". The vocabulary is 51 words.
- Transcripts come with the corpus, so no manual annotation is needed.
- We mix MUSAN noise into the audio at five levels: clean, 10, 5, 0 and minus 5 dB.
- We split by speaker, so test speakers are never seen in training.
- Limitation: it is studio-recorded, frontal and closed-vocabulary.

BACKGROUND YOU SHOULD UNDERSTAND

GRID was introduced by Cooke and colleagues in 2006 and is the standard starting corpus for sentence-level lip reading. Each of 34 speakers recorded 1,000 sentences, about 28 hours in total. Each clip is about 3 seconds, 75 frames at 25 frames per second.

Where 51 words comes from. Every sentence is command, colour, preposition, letter, digit, adverb. There are 4 commands, 4 colours, 4 prepositions, 25 letters (W is left out), 10 digits and 4 adverbs. 4 + 4 + 4 + 25 + 10 + 4 = 51. The letter slot is the hardest because many letters look alike on the lips.

Why this is good for a six-week project. The data exists now, it is already labelled, the audio and video are synchronised, and the faces are frontal and well lit, so mouth detection is reliable. Published work on GRID gives us reference points: LipSyncNet reports 8.2% WER on it [1].

MUSAN is a public corpus of music, speech and noise recordings made for exactly this purpose: mixing into clean speech to simulate noisy conditions. We scale the noise so that the mix has a chosen SNR. This gives a controlled experiment: the same sentence at five known noise levels.

Speaker-independent split. If the same person appears in training and test, the model can partly memorise that person's mouth and voice, and the test score looks better than it would on a new person. This is a form of data leakage. Splitting by speaker avoids it. The exact number of speakers in each split is a Week 1 decision.

Class balance. The grammar makes word frequency balanced within each slot, so imbalance is not a major concern, but we will confirm this in the Week 1 inspection.

NUMBERS ON THIS SLIDE AND WHERE THEY COME FROM

Number	What it is	Source
34	Speakers in GRID	LipSyncNet [1], proposal
~33,000	Usable sentences (34,000 recorded; some videos are missing)	Proposal; Afouras et al. [6]
51	Words in the vocabulary	Afouras et al. [6], Table 1
5	Noise levels: clean, 10, 5, 0 and minus 5 dB	Proposal, section III-A

Watch out: No sample frames are on this slide yet. The requirements ask for a sample of candidate data where possible. Add a strip of real GRID frames or mouth crops if you can.

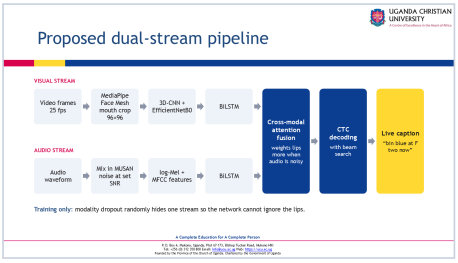
QUESTIONS YOU MAY BE ASKED

Show evidence that the data are actually obtainable.
GRID and MUSAN are public research corpora that can be downloaded today. Have a few real GRID frames and mouth crops on a laptop ready to show. The requirements warn that a proposal based on data the group has not located may be rejected.

Is 33,000 clips enough to train a deep network?
For a 51-word grammar, yes. Published models reach under 10% WER on GRID. We also start from a pretrained backbone, which reduces how much data is needed.

Why not use local data, for example Ugandan speakers or Luganda?
Collecting and labelling an audiovisual corpus is not feasible in six weeks. We will record a small set of consented clips of ourselves as an extra test, and we state plainly that GRID speakers are not representative of our intended users.

What are the licence terms?
Both are released for research use. Check the exact licence wording on each download page before the talk so you can quote it.



SLIDE 5 | 1:10

Proposed dual-stream pipeline

WHAT TO SAY

- Visual stream, left to right: video frames, then MediaPipe Face Mesh finds the lips and we crop the mouth to 96 by 96.
- A 3D-CNN with an EfficientNetB0 backbone extracts lip shape and motion. A BiLSTM models the sequence.
- Audio stream: the waveform with noise mixed in becomes log-Mel and MFCC features, encoded by its own BiLSTM.
- Fusion: cross-modal attention gives the lips more weight when the audio is noisy.
- CTC decoding with beam search turns the fused sequence into text.
- In training only, modality dropout hides one stream at random so the network cannot ignore the lips.

BACKGROUND YOU SHOULD UNDERSTAND

This is the slide with the most technical content. Each box is explained below in the order the data flows.

MediaPipe Face Mesh is a ready-made Google model that finds 468 three-dimensional landmark points on a face in real time from an ordinary camera. We take the landmarks around the lips, draw a tight box round them, crop and resize. We use it as a tool; we do not train it. It is more precise on small features like lips than the older Dlib detector, which gives only 68 points [3].

Why crop the mouth. The rest of the face and the background carry little speech information. Cropping removes distraction, makes the input small (96 by 96 instead of a full frame) and so cuts compute. The proposal allows 96 by 96 or 112 by 112; the slide shows 96.

3D-CNN. An ordinary (2D) convolution slides a filter over height and width of one image. A 3D convolution adds time: the filter covers a few consecutive frames at once. It therefore responds to movement, such as lips closing, and not only to shape.

EfficientNetB0 is a compact 2D image network pretrained on ImageNet. It is applied to each frame to give strong per-frame appearance features. In LipSyncNet the 3D-CNN features and the EfficientNetB0 features are joined together before the sequence model [1]. How exactly we combine the two is a design

detail to settle in Weeks 2 and 3.

BiLSTM. An LSTM is a recurrent network that reads a sequence step by step and keeps a memory. Bidirectional means two LSTMs, one reading forward and one backward, so each time step knows what came before and after. That context helps separate sounds that look the same in isolation.

log-Mel spectrogram and MFCC. Raw audio is a long list of pressure values. A spectrogram shows how much energy is at each frequency over time. The Mel scale spaces frequencies the way human hearing does. MFCCs are a compact summary of that spectrum, typically 13 numbers for each 25-millisecond window.

Cross-modal attention fusion. Simple fusion glues the two feature vectors together with fixed importance. Attention lets the network compute, at each moment, how much to rely on each stream. When the audio features look unreliable it shifts weight to the lips.

CTC (Connectionist Temporal Classification). We have 75 frames but only about six words, and no label saying which frame belongs to which letter. CTC solves this. The network outputs a character or a special blank at every frame; repeated characters are merged and blanks removed to give the text. Training adds up the probability of every frame-by-frame path that produces the correct sentence.

Beam search. At decoding time, greedy decoding takes the single most likely character at each frame. Beam search keeps several candidate sentences alive and picks the best overall, which usually lowers the error rate.

Modality dropout. Audio is the easier signal, so a fused network tends to lean on it and ignore the video. Randomly blanking one stream during training forces it to learn from both. Afouras and colleagues saw the same audio dominance and countered it by adding noise during training [6].

Watch out: Noise is mixed into the audio for training and for testing, so that we can measure each SNR level. Only modality dropout is training-only. Be clear about that if asked.

QUESTIONS YOU MAY BE ASKED

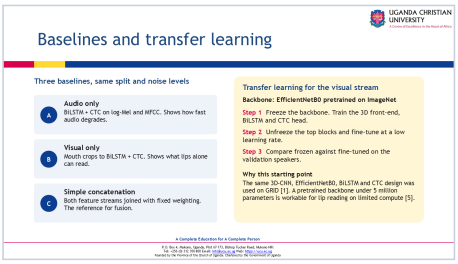
Why CTC and not a normal classification loss?
Cross-entropy needs one label per frame. GRID gives only the sentence. CTC learns the alignment itself.

Which parts do you train and which are off-the-shelf?
MediaPipe is used as it is. EfficientNetB0 starts pretrained and is fine-tuned. The 3D front-end, both BiLSTMs, the fusion module and the CTC output layer are trained by us on GRID.

Is it really live?
Not yet. A bidirectional LSTM needs the whole utterance before it can output, so captions appear per sentence after a short delay. True real-time operation is listed in our proposal as an open gap.

How does the network know the audio is noisy?
It is not told. It sees clean and noisy examples in training and learns that certain audio feature patterns are unreliable, then weights the lips more for those.

EfficientNetB0 expects colour ImageNet images. Your input is small mouth crops.
The network accepts other input sizes, and the early layers that detect edges and textures transfer well. We keep three-channel crops so the pretrained first layer can be reused. We will check this in the frozen-against-fine-tuned comparison.



SLIDE 6 | 0:50

Baselines and transfer learning

WHAT TO SAY

- Three baselines on the same split and noise levels: audio only, visual only, and simple concatenation.
- Each answers a question: how fast audio degrades, what lips alone can read, and what naive fusion gives.
- For transfer learning, the visual stream starts from EfficientNetB0 pretrained on ImageNet.
- Step 1, freeze the backbone and train around it. Step 2, unfreeze the top blocks at a low learning rate. Step 3, compare the two.
- LipSyncNet used this design on GRID, and a small pretrained backbone suits our limited compute.

BACKGROUND YOU SHOULD UNDERSTAND

Why a baseline is required. The requirements say a claim like "our model achieved 91%" is incomplete unless you can show what was achieved before, whether the comparison is fair, and what changed. Baselines are the reference for every later claim.

What each baseline tells us. Audio only shows how quickly speech recognition degrades with noise on our own data. Visual only shows the ceiling for lip reading alone. Concatenation shows what fusion gives without attention. If attention fusion does not beat concatenation, it has not earned its place.

Fair comparison means the same speakers in each split, the same preprocessing, the same noise levels and the same metric, changing one thing at a time.

Transfer learning. A network trained on a very large dataset has already learned general visual features: edges, textures, shapes. We reuse those weights instead of starting from random values. **ImageNet** is the standard source: over a million photographs in 1,000 classes.

Frozen and fine-tuned. Frozen means the pretrained weights are not updated; only the new layers learn. Fine-tuning means some pretrained layers are unfrozen and updated gently. We freeze first because the new layers start random, and their large early errors would otherwise damage the pretrained weights. The low learning rate in step 2 is for the same reason.

Why EfficientNetB0. It is small (about 5 million parameters) and accurate for its size, so it trains on modest hardware. LipSyncNet combined a 3D-CNN, EfficientNetB0, BiLSTM and CTC on GRID and reported 8.2% WER [1], so the design is proven on our dataset.

What reference [5] shows. The NASNetMobile study used a 4.7-million-parameter pretrained backbone for lip reading and found it beat larger ones such as VGG19. Note its accuracy was only 48% because it used a hard, uncontrolled dataset. It supports the choice of a small backbone, and it is also evidence for our domain-shift risk.

NUMBERS ON THIS SLIDE AND WHERE THEY COME FROM

Number	What it is	Source
8.2% WER	LipSyncNet on GRID, the design we follow	Jeevakumari and Dey [1]
4.7 million	Parameters in NASNetMobile, the "under 5 million" on the slide	Ameer et al. [5]

Watch out: EfficientNetB0 itself has about 5.3 million parameters. The "under 5 million" on the slide refers to the NASNetMobile backbone in reference [5]. Do not say EfficientNetB0 is under 5 million.

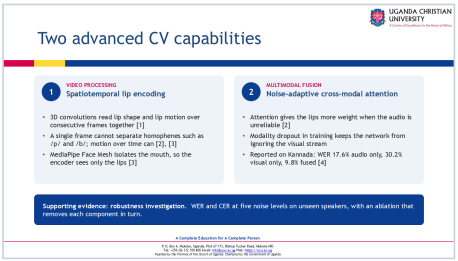
QUESTIONS YOU MAY BE ASKED

What is the baseline, and what will the final system be compared against?
Three baselines: audio only, visual only, and concatenation fusion. The final attention-fusion model is compared with all three at every noise level.

What will you do if the planned model does not improve over the baseline?
Report it as a finding. Use the error analysis to see where fusion fails, and submit the best baseline as the final system with the ablation as evidence. A negative result that is explained still meets the requirements.

Is this just a copy of LipSyncNet?
No. LipSyncNet is a lip-reading model. We add an audio stream, noise at controlled levels, attention fusion and modality dropout, and we evaluate robustness across noise levels, which that paper does not.

ImageNet is photographs of objects. Why would it help with lips?
The early layers learn general features such as edges and textures that apply to any image. The later layers are more specific, which is why those are the ones we unfreeze and adapt.



SLIDE 7 | 0:55

Two advanced CV capabilities

WHAT TO SAY

- Capability 1 is video processing: 3D convolutions read lip shape and motion over neighbouring frames.
- A single frame cannot separate homophenes like /p/ and /b/. Motion over time can.
- Capability 2 is multimodal fusion: attention that adapts to noise, with modality dropout in training.
- KanAVNet reports 17.6% word error rate for audio only, 30.2% for visual only and 9.8% fused.
- Supporting both, a robustness investigation across five noise levels on unseen speakers, with an ablation that removes each component in turn.

BACKGROUND YOU SHOULD UNDERSTAND

The rule. The requirements demand at least two advanced capabilities from a fixed list, and say each must have a clear role, not be added to look sophisticated. From that list we claim **video processing** and a **multimodal component**, with **robustness investigation** as a third.

Homophenes. Different sounds that look almost the same on the lips. /p/, /b/ and /m/ are all made by closing both lips, so "pat", "bat" and "mat" look alike. The difference is in the voice and inside the mouth, which a camera cannot see. Two things help: how the lips move over time, and the audio. That is the argument for both capabilities at once.

Why video processing counts as advanced. The system moves from isolated images to a sequence where time carries the meaning. The list describes video processing as extending the system "from isolated images to a meaningful video or repeated-frame task", which is exactly this.

Why fusion adds value. In KanAVNet, a similar CNN-BiLSTM-CTC system for Kannada, each stream alone was clearly worse than the two together: 17.6% and 30.2% alone, 9.8% fused [4]. Note that visual only is worse than audio only in clean conditions. Lips are the weaker signal on their own; their value is in

combination and in noise.

Ablation. Remove or switch off one component, retrain, and measure the change. If WER rises when the 3D front-end is replaced by a plain 2D one, the 3D front-end is contributing. It is the standard way to prove that each part matters.

Robustness investigation. Testing how the system behaves under realistic changes. Ours are noise level (five SNRs) and unseen speakers.

NUMBERS ON THIS SLIDE AND WHERE THEY COME FROM

Number	What it is	Source
17.6%	WER, audio only	KanAVNet [4]
30.2%	WER, visual only	KanAVNet [4]
9.8%	WER, audio and visual fused with attention	KanAVNet [4]

Watch out: The KanAVNet numbers are from a 25-speaker Kannada dataset built by its authors. Say "reported on Kannada" as the slide does. They are not GRID results and not ours.

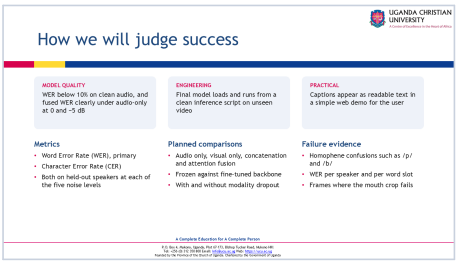
QUESTIONS YOU MAY BE ASKED

Why do these add value rather than complexity?
Each targets a named failure. The 3D encoder targets homophenes. The fusion targets noisy audio. The ablation will show with numbers whether each one earns its place.

The list describes the multimodal option as image and text. Yours is audio and video. Does it count?
We believe it does: it combines visual information with another modality to produce language. If it is not accepted, video processing and the robustness investigation are both on the list without doubt, so we still have two.

KanAVNet is Kannada on a custom dataset. Why is it relevant?
It is evidence for the pattern, not a target. A similar architecture on a different language showed fusion beating either stream. We will measure the same three-way comparison on GRID.

Why not use a transformer?
Transformers do well on large datasets such as LRS2 [6] but need more data and compute. For GRID's short, fixed sentences a BiLSTM is sufficient and cheaper. It is a reasonable later comparison.



SLIDE 8 | 0:50

How we will judge success

WHAT TO SAY

- Our primary metric is word error rate, with character error rate, on held-out speakers at every noise level.
- Model quality: below 10% WER on clean audio, and fused clearly better than audio only at 0 and minus 5 dB.
- Engineering: the final model loads and runs from a clean inference script on unseen video.
- Practical: captions appear as readable text in a simple web demo.
- We compare the three baselines with attention fusion, frozen with fine-tuned, and with and without modality dropout.
- For failures we look at homophene confusions, error per speaker and per word slot, and frames where the mouth crop fails.
- Targets may be revised after the baseline, and we will explain any change.

BACKGROUND YOU SHOULD UNDERSTAND

Why WER and not accuracy. Accuracy suits one label per input. Here the output is a sentence that can be partly right. WER counts how many words are wrong and is the standard metric in speech recognition.

Character error rate (CER) is the same calculation on characters. It is more forgiving and more detailed: "bin" read as "pin" is one whole word error but only one character in three. Reporting both shows whether errors are near-misses or complete failures.

Three kinds of success criteria are required: model quality, engineering and practical. The three cards on the slide map to those one for one.

Why below 10%. Published GRID systems report about 8 to 10% WER (LipSyncNet 8.2% [1]), so the target is realistic but not trivial. The requirements allow targets to be revised after the baseline as long as the change is explained and not made silently.

Controlled comparison. Change one thing, keep everything else fixed. That is how we know which change caused an improvement.

Failure analysis. The requirements want inspection of actual predictions, not only a score. A confusion breakdown shows which words or letters get mixed up. We expect the letter slot and homophene pairs to dominate. Error per speaker shows whether some faces are harder. Counting frames where MediaPipe loses the mouth separates detection failures from recognition failures.

Clean inference script. One script that takes a video file and prints a caption, with no dependence on notebook state or paths on one person's laptop. The requirements name this explicitly.

NUMBERS ON THIS SLIDE AND WHERE THEY COME FROM

Number	What it is	Source
below 10%	Target WER on clean audio for the fused model	Proposal, section IV
0 and minus 5 dB	Noise levels where fusion must clearly beat audio only	Proposal, section IV

QUESTIONS YOU MAY BE ASKED

What metric will matter most, and what does it measure?

Word error rate: substitutions, deletions and insertions divided by the number of words in the correct sentence. Lower is better.

What does "clearly better" mean? Give a number.

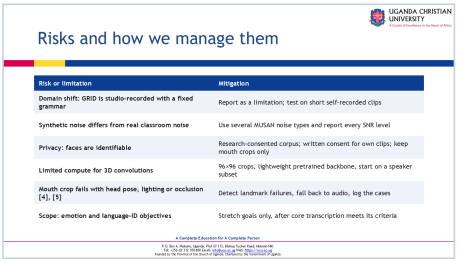
We will set the number after the Week 2 baseline, as the requirements allow. From the literature, a drop of about ten percentage points at 0 dB (32.5% to 22.4% in [2]) is a realistic expectation.

How do you know the model is not just memorising the grammar?

It partly can, and that is a known property of GRID. That is why we report per-slot error: the letter and digit slots cannot be guessed from grammar, so they show real recognition ability.

What is the most dangerous error this system could make?

A fluent but wrong caption that the user trusts, such as a swapped letter or digit. A missing word is visible; a wrong one is not. That is why the system supports the user and does not replace human review.



SLIDE 9 | 0:45

Risks and how we manage them

WHAT TO SAY

- Domain shift: GRID is studio-recorded with a fixed grammar. We state it as a limitation and test on short clips of our own.
- Synthetic noise is not real classroom noise. We use several noise types and report every level.
- Privacy: faces are identifiable. Our own clips need written consent, and we keep mouth crops only.
- Compute: small crops, a lightweight pretrained backbone, and a speaker subset first.
- The mouth crop can fail with pose, lighting or occlusion. We detect it, fall back to audio and log the cases.
- Scope: emotion and language identification are stretch goals, only after core transcription meets its criteria.

BACKGROUND YOU SHOULD UNDERSTAND

The requirements ask for **at least five** meaningful risks. We give six. Do not read the table out; pick the two or three you can speak about best and point to the rest.

Domain shift means the conditions at use differ from the conditions in training. GRID is studio lighting, one frontal speaker, a 51-word grammar. A lecture room has varied lighting, moving heads and open vocabulary. Reference [5] shows what happens on uncontrolled data: 48% accuracy. This is the biggest limitation of the project, so name it first.

Synthetic noise. We add recorded noise to clean speech by computer. Real rooms also have echo, and people speak differently when it is loud. So our noise results are an approximation.

Privacy. The requirements say that where people are identifiable, consent, privacy, data storage and appropriate use must be addressed. A system that watches faces continuously is sensitive. Our position: GRID was recorded for research with its participants' agreement; any clips of ourselves need written consent; we store mouth crops, not full faces; and the system does not identify anyone.

Compute. 3D convolutions over 75 frames are expensive in memory and time. Small crops, a small backbone and starting on a subset of speakers keep experiments short enough to iterate.

Occlusion and pose. Hands, masks, facial hair or a turned head can make MediaPipe lose the lips [4], [5]. If no mouth is found, the system should rely on audio and record the event, not guess.

Scope. The requirements list "unrealistic scope" as a pattern that will not be approved. Emotion detection and language identification are separate tasks needing different data, so we make them stretch goals. Expect to be asked about this, since they are in the proposal's objectives.

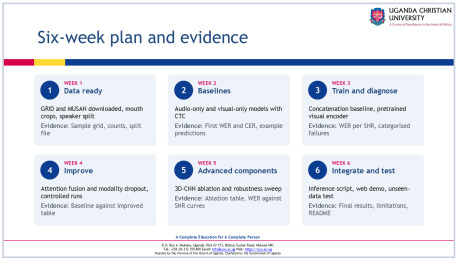
QUESTIONS YOU MAY BE ASKED

Your objectives include emotion detection and language identification. Where are they in the plan?
They are stretch goals. GRID has neutral speech in one language, so it cannot train either. We chose to do robust transcription properly in six weeks and attempt the others only if the core criteria are met.

How will you handle consent for your own recordings?
Only group members will be recorded, each with written consent. Clips stay on our own machines, are reduced to mouth crops, and are not published.

Could this be misused to read what people say from a distance?
The same technique could be. That is why the prototype needs a close frontal view, does not identify speakers, and is framed as an aid used with the speaker's knowledge. We will state this in the report.

What if you run out of compute?
Train on fewer speakers, reduce the crop size, and keep the backbone frozen. All three reduce cost sharply. We would report the reduced setting.



SLIDE 10 | 0:40

Six-week plan and evidence

WHAT TO SAY

- Week 1, data: GRID and MUSAN downloaded, mouth crops, speaker split. Evidence: sample grid, counts, split file.
- Week 2, baselines: audio only and visual only. Evidence: first WER and CER, example predictions.
- Week 3, train and diagnose: concatenation baseline and pretrained visual encoder. Evidence: WER per noise level, categorised failures.
- Week 4, improve: attention fusion and modality dropout. Evidence: baseline against improved table.
- Week 5, advanced components: 3D-CNN ablation and robustness sweep. Evidence: ablation table and WER against noise curves.
- Week 6, integrate and test: inference script, web demo, unseen-data test. Evidence: final results, limitations, README.
- Every member can explain every stage. Thank you, we welcome questions.

BACKGROUND YOU SHOULD UNDERSTAND

The plan follows the week-by-week table in the requirements, which names a **main target** and **minimum evidence** for each week. Each of our weeks produces something that can be shown.

Do not read all six cards. Say the shape: data, baselines, diagnose, improve, advanced components, integrate. Then give one example of evidence.

Why this order. Baselines come before the full model so that every later change has a reference. Error analysis in Week 3 comes before improvement in Week 4 because the requirements want improvements motivated by observed failures, not guessed.

The highest-risk week is Week 1. If mouth cropping across 33,000 videos or the noise mixing is late, everything slips. Start the downloads before the proposal is approved.

Keep evidence from day one. The final report must reconstruct the whole journey, including failed attempts. Save training curves, experiment tables and example errors as they happen.

Reproducibility. The requirements expect a repository with a README, separated code for data preparation, training, evaluation and inference, model weights, results and a data statement.

QUESTIONS YOU MAY BE ASKED

How will another person reproduce your final inference pipeline?

A repository with a README, a fixed split file, saved weights and one inference script that runs on a new video with no notebook state or personal paths.

Is this achievable in six weeks?

The dataset is ready-made and labelled, the architecture follows published work, and scope is limited to transcription. The riskiest part is compute, which we manage by starting small.

Who is doing what?

Work is divided, but the requirements are clear that knowledge is not. Agree the split of tasks before the talk and be ready to say it, and make sure each person can explain the others' parts.

How did you use AI tools?

AI use is allowed, but you may not submit anything you cannot explain, modify and defend. Answer honestly about what was used and how you checked it. The final report needs an AI Assistance Statement.

Glossary

Short definitions to revise from. Each is explained more fully under the slide noted.

Term	Meaning	Slide
3D-CNN	Convolution over time as well as height and width; captures motion.	5
Ablation	Removing one component and re-measuring, to show what it contributes.	7
ASR	Automatic speech recognition: audio to text.	2
Attention	A mechanism that learns how much weight to give each input at each moment.	5
AVSR	Audio-visual speech recognition: sound and video together.	3
Babble noise	Background noise made of many people talking.	2
Baseline	The simplest reasonable system, used as the reference for later changes.	6
Beam search	Decoding that keeps several candidate sentences and picks the best overall.	5
BiLSTM	Recurrent network that reads a sequence forward and backward.	5
CER	Character error rate: WER computed on characters.	8
CTC	Loss that maps many frames to a shorter text without frame-level labels.	5
Domain shift	Use conditions differ from training conditions.	9
EfficientNetB0	Compact image network pretrained on ImageNet; our visual backbone.	6
Fine-tuning	Unfreezing pretrained layers and updating them gently on our data.	6
GRID	Audiovisual corpus: 34 speakers, fixed six-word sentences, 51 words.	4
Homophenes	Different sounds that look the same on the lips, such as /p/, /b/, /m/.	7
ImageNet	Large photograph dataset (1,000 classes) used to pretrain image networks.	6
MediaPipe Face Mesh	Ready-made model giving 468 facial landmark points in real time.	5
MFCC / log-Mel	Compact audio features describing the sound spectrum over time.	5
Modality dropout	Hiding one input stream at random in training so neither is ignored.	5
MUSAN	Public corpus of music, speech and noise used to add noise to clean audio.	4
SNR	Signal-to-noise ratio in dB. 0 dB means speech and noise are equally strong.	2
Speaker-independent	Test speakers never appear in training.	4
Transfer learning	Starting from weights learned on another, larger dataset.	6
VSR	Visual speech recognition: lip reading by computer.	3
WER	Word error rate: (substitutions + deletions + insertions) / words. Lower is better.	2

REFERENCE NUMBERS USED IN THIS GUIDE

[1] Jeevakumari and Dey, LipSyncNet, IEEE Access 2024. [2] Li et al., Sensors 2023. [3] Aripin and Setiawan, 2024. [4] Divya and Suresha, KanAVNet, 2026. [5] Ameer et al., NASNetMobile, IEEE Access 2025. [6] Afouras et al., IEEE TPAMI 2022. [9] Kumar et al., 2022. Full details are on slide 11.